

Unit Test Summary

Test Framework

- **Framework:** Jest v29.7.0
- **HTTP Testing:** Supertest v7.2.2
- **Approach:** Unit testing with mocked models (no live database required)
- **Run Command:** npm test

Run Command

1. __tests__/auth.test.js - User Authentication

#	Test Case	Type	Expected Result
1	Successful registration redirects to login	Success	302 redirect to /login to
2	Successful login creates a session and redirects to profile	Success	302 redirect to /profile
3	Failed login (wrong password) redirects back to login	Failure	302 redirect to /login

2. __tests__/flightController.test.js - Flight Management

#	Test Case	Type	Expected Result
1	Creates a flight and redirects on success	Success	Redirect to /admin/flights, audit logged
2	Rejects when flight number already exists	Failure	400 error
3	Rejects when departure is not before arrival	Failure	400 error
4	Updates a flight and returns success JSON	Success	200 with { success: true }
5	Returns 400 when required fields are missing (update)	Failure	400 error
6	Returns 404 when flight to update is not found	Failure	404 error
7	Deletes a flight and returns success JSON	Success	200 with { success: true }, audit logged

#	Test Case	Type	Expected Result
8	Returns 404 when flight to delete is not found	Failure	404 error

3. `__tests__/reservationController.test.js` - Reservation Management

#	Test Case	Type	Expected Result
1	Creates a reservation, decrements seats, and redirects	Success	Redirect to /admin/reservations, audit logged
2	Rejects reservation when flight number does not exist	Failure	404 error
3	Rejects reservation when flight has no available seats (business rule)	Business Rule	400 error — "Only 0 seat(s) left"
4	Rejects reservation when selected seat is already taken (business rule)	Business Rule	400 error — "already taken"
5	Cancels a reservation, releases the seat, and returns success	Success	200, status changed to Cancelled, seat restored
6	Returns 404 when reservation to cancel is not found	Failure	404 error
7	Rejects cancelling a reservation that is already cancelled	Failure	400 error
8	Rejects an invalid reservation ID format	Failure	400 error

4. `__tests__/bookingController.test.js` - Booking (Passenger-Side)

#	Test Case	Type	Expected Result
1	Rejects booking when the flight has no available seats (business rule)	Business Rule	Renders booking page with error
2	Rejects booking when the selected seat is already taken (business rule)	Business Rule	Renders booking page with error
3	Successfully books a flight when seat is available and unoccupied	Success	Redirect to /my-reservations, audit logged
4	Rejects booking when required passenger fields are missing	Failure	Renders booking page with validation errors

Test Summary

Category	Total Tests	Success Cases	Failure Cases	Business Rule Tests
User Authentication	3	2	1	0
Flight Management	8	3	5	0
Reservation Management	8	2	6	2
Booking (Passenger)	4	1	3	2
Total	23	8	15	4

Business Rule Validations Covered

1. **Booking a flight with no available seats.** Tests that the system rejects bookings when availableSeats is 0
2. **Selecting an occupied seat.** Tests that the system prevents double-booking of the same seat on the same flight
3. **Creating a reservation with no seats left.** Validates the admin creation path also enforces seat availability.
4. **Creating a reservation with a taken seat.** Validates the admin creation path also checks for seat conflicts.